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Improved LSTM hyperparameters alongside sentiment walk-forward validation for time series prediction

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ABSTRACT

This study aims to address the common issue of biased estimation errors in time series modeling by analyzing the error in locating ideal hyperparameters and defining appropriate validation methods. Specifically, it focuses on predicting the stock price of Bank Rakyat Indonesia using a combination of historical stock prices, technical indicators, exchange rates, and news sentiment data, while determining the optimal variables for deep learning models. Employing a deep learning-based Long-Short Term Memory (LSTM) model, the study optimizes hyperparameters alongside walk-forward validation for time series prediction. It explores different combinations of variables and adapts the sliding window approach to the context of the data. The results highlight the importance of optimizing hyperparameters and utilizing walk-forward validation for accurate time series prediction. The model incorporating historical stock prices and sentiment scores outperforms others, achieving an RMSE of 96.61 and MAE of 86.97. Incorporating sentiment scores reduces RMSE by 39.55 % compared to models using only historical stock prices, while adding technical indicators does not yield improvement. This study contributes to the field by addressing the issue of biased estimation errors in time series modeling, offering insights into the optimization of hyperparameters and validation methods for accurate predictions. It also underscores the significance of incorporating sentiment analysis alongside historical stock prices for improved forecasting accuracy.

1. Introduction

Stock prices are dynamic and adapt quickly to changes in the monetary sector (Biswas, 2020). Stock prices are influenced by many macroeconomic factors such as political activities, general economic conditions, company policies, expectations, and the psychology of investors (Tan et al., 2007). This makes stock price prediction challenging. Even so, in the academic realm, the topic of stock price prediction is

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widely studied and the number continues to increase from year to year (Bustos and Pomares-Quimbaya, 2020). The techniques used are also varied, ranging from the use of statistical-based models such as Autoregressive Integrated Moving Average (ARIMA) (Ariyo et al., 2014) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) (Mustapa and Ismail, 2019), machine learning-based models such as Support Vector Machine (SVM) (Ardyanta and Sari, 2021) to using deep learning-based models such as Long Short Term Memory (LSTM) (Heiden and Parpinelli, 2021; Wu et al., 2022)

In addition, there are various independent variables used. Research conducted by (Kara et al., 2011; Sivapurapu, 2020) uses technical indicators to predict stock prices. In addition, some use fundamental data as an independent variable. Fundamental data includes company profile data, financial conditions, market conditions, political, business, and economic climate (Ardyanta and Sari, 2021). However, fundamental data is often reported with low frequency, the reporting date is not accurate, and it is only reported within a certain time so it is not useful for use in daily stock price predictions (Jiang, 2021). Fundamental data that is often used is currency exchange rates because, like stock prices, currency exchange rate data is also available every day. Currency exchange rates have been used in research (Ardyanta and Sari, 2021) and (Zhong and Enke, 2017).

In addition, the sentiment of text data can also be used as an independent variable (Derakhshan and Beigy, 2019). The text data can come from social media (Nti et al., 2020; Ruan et al., 2018) and news portals (Heiden and Parpinelli, 2021; Mohan et al., 2019). The amount of text data circulating on the internet can affect investor sentiment which will have an impact on changes in stock prices (Nguyen et al., 2015). When investor sentiment is negative it can cause stock prices to decline, and vice versa when investor sentiment is positive it can cause stock prices to rise (Nti et al., 2020).

With the many variables available, some researchers are curious to learn which ones should be used most effectively. As done by Heiden and Parpinelli (2021) and Nti et al. (2020), they included the news score sentiment variable as an independent variable. The results obtained indicate that the model with the sentiment score has better accuracy than the model that uses the history of stock prices. However, both studies still have 3 shortcomings that can cause the results obtained to be biased.

The first is that the model used has the same hyperparameter configuration on different data. For example in research (Heiden and Parpinelli, 2021) using a 2-layer LSTM with the first layer having 64 units and the second layer having 32 units. The configuration is applied to 2 different data combinations. The first combination only uses 1 independent variable, namely stock price history (M_1) and the second uses 2 independent variables, namely stock price history, and financial news sentiment data (M_2). Whereas according to Kohavi and John (1995), each different hyperparameter configuration tends to be good for different data sets. Therefore, it is possible that the combination of variables M_1 and M_2 has different hyperparameter configurations to achieve maximum performance. It is only reasonable to compare the performance of the two combinations of variables after reaching maximum performance.

The second is the determination of the hyperparameters used without performing hyperparameter optimization (HPO). Whereas according to Victoria and Maragatham (2021) the selection of hyperparameters plays an important role in obtaining a model with good performance. It is possible that the hyperparameter determined by the researcher is not the best combination of hyperparameters for the combination of M_1 and M_2.

The third is the use of holdout validation, in which the data is divided into training and testing data. The use of holdouts will result in the reported error, which may only represent an error in the testing data, not an error in the data as a whole (Bergmeir and Benítez, 2012). Because the data used is time series data, the method used is walk-forward validation so that the relationship between time is also considered.

These three shortcomings can result in the reported error not being the minimum error that can be obtained by the model. This will result in incorrect conclusions about which combination of variables is appropriate for predicting stock prices. Thus, the motivation of this research is to fill in the shortcomings of the two previous studies.

The main contribution of this research is to present a general framework for modeling time series data. Although this study uses stock price data, the framework presented can be generalized to other time series data. The model used in this study is an LSTM which has been proven to be able to study sequential data and overcome the problem of exploding and vanishing gradients in ordinary Recurrent Neural Network (RNN) (Graves et al., 2009; Hochreiter and Schmidhuber, 1997; Mndawe et al., 2022).

In this study, HPO will be conducted to determine the best model for each combination of variables. Each combination of variables will then be compared with the walk-forward validation method. The use of this validation method will simulate what happens in the stock market, where stock data will increase by one every day (Chen, Jiang, et al., 2023). So that the reported error can describe the out-of-sample error more accurately because the model already has the best hyperparameters and is validated according to the actual situation in the stock market.

This study uses historical closing price data, technical indicators, currency exchange rates, and news sentiment data. To the best of our knowledge, this study is the first study to use these four types of variables simultaneously, accompanied by HPO and walk-forward validation. From the four types of variables, 8 combinations of variables will be formed which will be compared with each other. So that the final result of this study is a model with a combination of variables that provides the best accuracy in predicting stock prices.

The remainder of this paper is as follows. Section 2 contains related research that also predicts stock prices. Section 3 contains the methodology used in this study. Section 4 contains a discussion and experimental results obtained. Finally, conclusions and suggestions for further research are summarized in Section 5.

2. Related studies

Table 1 shows a literature review of 16 scientific articles that predict stock prices or movements. All studies use stock price history as an independent variable. Only research (Ardyanta and Sari, 2021) uses 4 variables simultaneously, but neither HPO nor walk-forward validation is used. The study determined the hyperparameters used and only used the holdout validation method with 70 % training data and 30 % testing data.

There are 2 studies that do HPO but do not use walk-forward validation (Kara et al., 2011; Mohan et al., 2019). For example, research (Mohan et al., 2019) conducted HPO manually with a trial and error process but only used the holdout validation method. slightly different from what was done by Ardyanta and Sari (2021), Mohan et al divided the data into 3 parts, namely 90 % training data, 5 % validation data, and 5 % testing data. In the HPO process, the training and validation data are both utilized.

Research (Kara et al., 2011) uses a grid search algorithm to perform HPO. However, the study divides the data by considering the number of days the stock price goes up and down. For example, in 2006 training data where the stock price went up by 13 days and decreased by 12 days, then in the testing data, the stock price went up by 13 days and down by 12 days. This method is not appropriate for use on stock price data which is time series data. Although the proportion of training and testing data has been considered, the daily data dependency is lost and is not considered.

Six studies have conducted HPO and used walk-forward validation (Börjesson and Singull, 2020; Huang et al., 2018; Kouassi and Moodley, 2020b, 2020a; Siami-Namini et al., 2018; Sivapurapu, 2020). However, this study does not use stock price history variables, sentiment scores,

Table 1
Literature study results.

Reference	Data Usage				Hyperparameter Optimization	Walk Forward Validation
	History	Technical Indicator	Exchange Rate	Sentiment		
(Heiden and Parpinelli, 2021; Nguyen et al., 2015; Nti et al., 2020)	✓	×	×	✓	×	×
(Mohan et al., 2019)	✓	×	×	✓	Manual	×
(Mndawe et al., 2022; Wu et al., 2022)	✓	✓	×	✓	×	×
(Deng et al., 2011)	✓	✓	×	✓	×	✓
(Ardyanta and Sari, 2021)	✓	✓	✓	✓	×	×
(Zhong and Enke, 2017)	✓	✓	✓	×	×	×
(Börjesson and Singull, 2020; Siami-Namini et al., 2018)	✓	×	×	×	Manual	✓
(Kouassi and Moodley, 2020b, 2020a)	✓	×	×	×	Bayesian Optimization and HyperBand, Manual	✓
(Sivapurapu, 2020)	✓	✓	×	×	Genetic Algorithm, Tree-structured Parzen Estimator Approach	✓
(Kara et al., 2011)	✓	✓	×	×	Grid Search	×
(Huang et al., 2018)	✓	×	×	×	Bayesian Optimization	✓
This study	✓	✓	✓	✓	Bayesian Optimization	✓

technical indicators, and exchange rates simultaneously. The research also does not focus on determining which variables are the best to use. As done by Kouassi and Moodley (2020b), this research focuses on discussing the performance of the Bayesian Optimization and HyperBand (BOHB) algorithm to determine the best hyperparameters of the LSTM model used in time series data. So that the variable used is only the history of stock prices and does not compare it at all with the use of other variables. Similar to research (Sivapurapu, 2020), Similar to research [10], which compares the effects of the variables used with the usage of various statistical and deep learning-based models.

3. Materials and methods

In this section, the general framework proposed for modeling time series data will be explained. In general, the framework is described in Fig. 1.

3.1. Data collection

The data used in this study is secondary data. There are 3 main data used, namely historical data on Bank Rakyat Indonesia (BRI) stock prices, the rupiah exchange rate, and news articles. The three data will be further processed to obtain 13 independent variables that will be used. Stock price history data was obtained from Yahoo Finance, rupiah

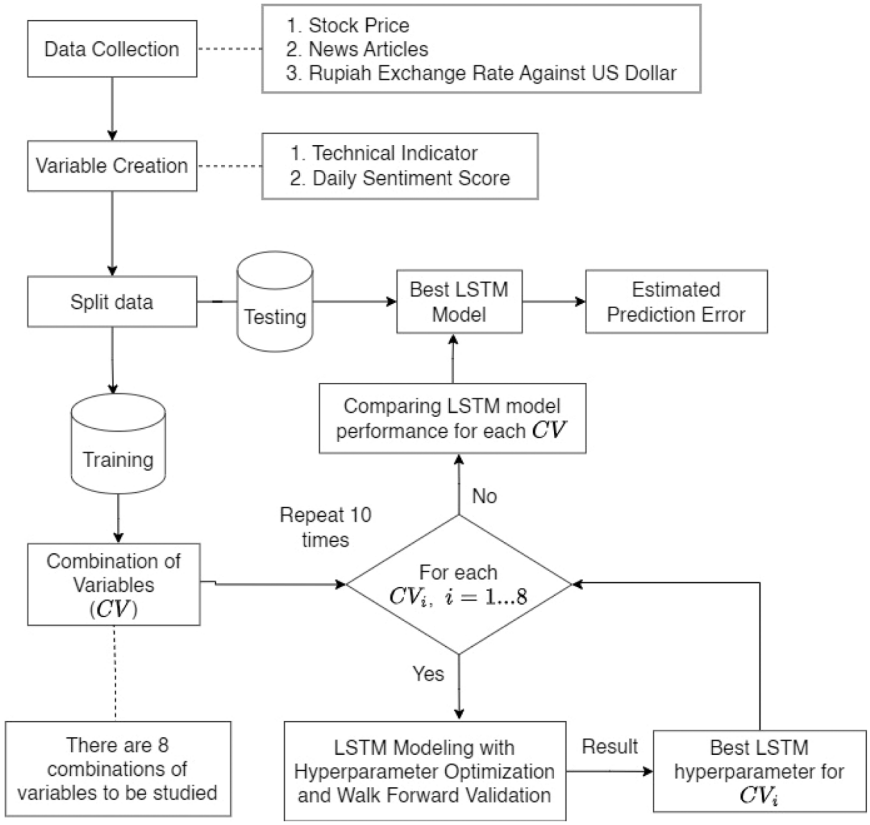


Fig. 1. Research Flow.

exchange rate data were obtained from Investing.com, and article data was obtained from detik finance.com and Bisnis.com.

The article data obtained still needs to be filtered first to retrieve articles that are only related to BRI. The keywords used are the company name "Bank Rakyat Indonesia", "BRI" and the BRI stock code, namely "BBRI". The article that will be taken is an article whose title or content contains one of these keywords.

3.2. Variable creation

3.2.1. Technical indicators

Technical indicators are calculated using the closing price, opening price, highest price, lowest price, and selling volume (Ardyanta and Sari, 2021; Sivapurapu, 2020). In this study, 10 technical indicators will be used which were selected from previous studies. Table 2 describes the mathematical formula of the selected technical indicators.

3.2.2. Sentiment score

Daily score sentiment will be calculated using the method that has been done in research (Heiden and Parpinelli, 2021). The study conducted sentiment analysis with a lexical-based approach using the Valence Aware Dictionary for Sentiment Reasoning (VADER) model (Hutto and Gilbert, 2014). Fig. 2 shows the process for obtaining daily score sentiment.

Articles that have been collected will be preprocessed first. We removed unnecessary sentences containing advertisements which are often found at the end of articles. In addition, punctuation corrections were also carried out, such as removing periods that did not mark the end of a sentence. Articles are not converted to lowercase (case folding) because the VADER model can detect differences in the use of capital letters or not and can give different results (Hutto and Gilbert, 2014). For example, the sentence "I'AM HAPPY" will have a more positive sentiment than the sentence "I'm happy".

Because VADER uses a dictionary of words in English, the article will be translated into English first. After being translated, each article is analyzed using VADER and will produce a score in the range of −1–1. The closer to −1, the article is considered to have a negative sentiment and vice versa.

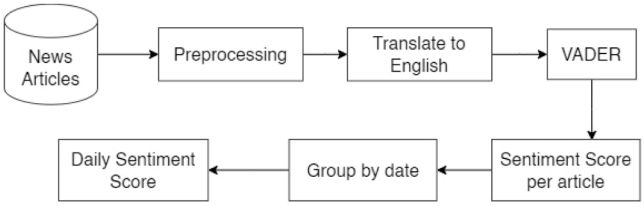


Fig. 2. The flow of getting daily score sentiment.

The daily sentiment score is obtained by averaging all the sentiment scores from the articles on that day. However, because the stock market is not open every day, the sentiment scores on the day the stock market is not open will be aggregated to the next day when the stock market is open. For example, because Saturday and Sunday the stock market is closed, Monday's sentiment score is the average sentiment score of Saturday, Sunday, and Monday articles. Days without articles will have a score of 0.

3.3. Splitting data and combining variables

The data will be split into training and testing data. Training data starts from 15 December 2003 to 31 December 2021. Training data will be used to determine the best hyperparameters for each combination of variables, model comparisons, and comparisons of variables combinations.

Meanwhile, data testing will start from early January 2022 until the end of March 2022. Data testing should not be used in the modeling process so that information leakage or data leakage does not occur (Zheng and Casari, 2018). Test data will only be used when the modeling

Table 2
Selected Technical Indicators and Their Formulations.

Indicator Name	Ever Used in Research				Formula
	(Kara et al., 2011)	(Ardyanta and Sari, 2021)	(Sivapurapu, 2020)	(Vargas et al., 2018)	
Simple 10-day Moving Average (SMA)	✓	✓	✓	×	$\frac{C_t + C_{t-1} + \ldots + C_{t-10}}{10}$
Weighted 10-day Moving Average (WMA)	✓	×	✓	×	$\frac{10C_t + 9C_{t-1} + \ldots + C_{t-10}}{10 + 9 + \ldots + 1}$
Moving Average Convergence Divergence	✓	×	✓	✓	$EMA_{12}(t) - EMA_{26}(t)$
10-day Momentum	✓	×	✓	✓	$C_t - C_{t-10}$
Stochastic %K	✓	×	✓	✓	$\frac{C_t - LL_{t-14}}{HH_{t-14} - LL_{t-14}} \times 100$
Stochastic %D	✓	×	✓	✓	$\frac{\sum_{i=0}^{2} K_{t-i}\%}{3}$
Relative Strength Index	✓	✓	✓	✓	$100 - \frac{100}{1 + \left(\frac{\sum_0^{13} Up_{t-i}}{14}\right) + \left(\frac{\sum_{i=0}^{13} Dw_{t-1}}{14}\right)}$
Larry William's R%	✓	×	✓	✓	$\frac{HH_{t-14} - C_t}{HH_{t-14} - LL_{14}}$
Commodity Channel Index	✓	×	✓	×	$\frac{M_t - SM_t}{0.015D_t}$
Money Flow Index	×	✓	×	×	$100 - \frac{100}{1 + \frac{14-day\ positive\ MF}{14-day\ negative\ MF}}$

Notes: C_t is the closing price, L_t is the lowest price, dan H_t is the highest price at time t. LL_{t-14} is the lowest price from L_{t-14} to L_t, HH_{t-14} is the highest price from H_{t-14} to H_t. EMA is Exponential Moving Average, $EMA_k(t) = EMA_k(t-1) + \alpha \times (C_t - EMA_k(t-1))$, with α is smoothing factor $= \frac{2}{1+k}; k = 14$. $M_t = \frac{H_t + L_t + C_t}{3}$; $SM_t = \frac{\sum_{i=1}^{20} M_{t-i+1}}{20}$; $D_t = \frac{1}{20}\sum_{i=1}^{20} |M_{t-i+1} - SM_t|$; MF is money flow. $MF = M_t \times transaction\ volume$. Up_t means that the price change goes up and Dw_t means that the price change goes down at time t.

process has been completed. The data is used to estimate the prediction error of the best model obtained. If testing data is used in the modeling process, the estimated prediction error obtained will be biased (Kuhn and Johnson, 2020).

Because the variable that will be the target in the modeling is the closing price, the historical closing price variable will always be used. In addition, 10 technical indicator variables will be included in the same group, so we can see the effect of including technical indicators. Therefore, in this study, there are 8 combinations of variables to be studied, which are described in Table 3.

3.4. Stock price prediction modeling

The model is made to predict stock prices in the next 5 days so that it is included in the multi-step time series case. The model that will be used is LSTM because the model has been proven to apply to time series data and has been used in many studies. HPO and 10-fold walk-forward validation were used to obtain the LSTM model with the best performance for each combination of variables. The evaluation metrics used are Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). This process will be repeated 10 times for each combination of variables so that it can see the stability of the HPO algorithm used.

3.4.1. Long Short-Term Memory (LSTM)

LSTM is a type of Recurrent Neural Network (RNN) that can study sequential data and solve the problem of exploding and vanishing gradients in ordinary RNN (Graves et al., 2009; Hochreiter and Schmidhuber, 1997; Mndawe et al., 2022). Information on the LSTM is stored in a memory cell which is managed through three gates, namely, the input gate, forgot gate, and the output gate (Siami-Namini et al., 2018; Wu et al., 2022).

In this study, the model used will have a maximum 3-layer LSTM. Between each layer will be added a dropout layer to avoid overfitting (Srivastava et al., 2014). Because the model predicts the next 5 days, the output layer is a dense layer with a total of 5 units.

The input layer in the LSTM has a three-dimensional size (Brownlee, 2018). The first dimension is to determine the amount of data to be used, the second dimension is to determine the number of previous observations that will be used (Aslam et al., 2024), and the third dimension is to determine the number of variables to be used (Chen et al., 2022; Chen, Niu, et al., 2023; Ji et al., 2025). The input layer in an LSTM is usually denoted by $(samples, timestamps, features)$ (Caraka et al., 2024; Toharudin et al., 2021). In this study the features will depend on the combination of variables used, for example for VC_8, the features value is 13 because there are 13 variables used. Meanwhile, the value of timestamps will be determined in the HPO process because it cannot be determined with certainty how many previous observations are best used to predict the next 5 days.

Table 3
A combination of variables used in this study.

Variable Combination (VC)	Variable
VC_1	Closing price history
VC_2	History of closing prices and sentiment scores
VC_3	History of closing prices and rupiah exchange rate
VC_4	History of closing prices and 10 technical indicators
VC_5	History of closing prices, sentiment scores, and the rupiah exchange rate
VC_6	History of closing prices, sentiment scores, and 10 technical indicators
VC_7	History of closing prices, rupiah exchange rates, and 10 technical indicators
VC_8	History of closing prices, score sentiment, rupiah exchange rate, and 10 technical indicators

3.4.2. Hyperparameter optimization

In this study, HPO uses a probabilistic search approach (Torres et al., 2021), namely Bayesian Optimization. This method will consider the previous evaluations that have been done to determine the next hyperparameter to be evaluated. This is the main advantage of Bayesian Optimization compared to other algorithms that are not adaptive, such as grid search and random search (Bergstra and Bengio, 2012).

Bayesian Optimization has two important things, namely the surrogate model to model the objective function and the acquisition function to determine the next hyperparameter to be evaluated (Yu and Zhu, 2020). The objective function used in this study is the average RMSE of 10-fold walk-forward validation and the budget used is the number of iterations of 50. This means that for each run, Bayesian Optimization will look for 50 hyperparameter configurations for each variable combination. Table 4 describes the hyperparameters that will be optimized in this study along with their limitations.

The number of epochs is not included in the HPO process because during the training process the model will use Early Stopping (ES) with a maximum number of epochs of 100. The activation function is also not included in the HPO process and only uses the hyperbolic tangent (tanh) according to the original model (Hochreiter and Schmidhuber, 1997). Activation functions such as Rectified Linear Unit (ReLu) are not recommended for use in LSTMs unless accompanied by certain weight initialization techniques (Le et al., 2015).

3.4.3. 10-fold walk-forward validation

Randomization of data using cross-validation is not appropriate when used on time series data that is dependent on time. It is quite strange to use future data to predict past data (Börjesson and Singull, 2020; Kouassi and Moodley, 2020a). Therefore, the walk-forward validation method is suitable for use on time series data because the dependence between t_n and t_{n+1} data is considered (Barra et al., 2020). An illustration of walk-forward validation is presented in Figure 3.

Because the modeling process uses early stopping, at each fold, the data will be divided into 3, namely the training set, early stopping set, and test set. The data on the training set will be standardized using the Min-Max Scaler. The data that will be monitored by early stopping must be distinguished from the data that will be used to evaluate the model so that data leakage can be avoided (Kuhn and Johnson, 2020). When the RMSE of the early stopping set increases 3 times in a row, the model training process will be stopped (Prechelt, 1998). When the training process is complete, the model will be returned to the best epoch which has the smallest RMSE in early stopping set. Then the test set will be evaluated on the model to get the RMSE of the fold.

In addition, ReduceLRonPlateau is also used with patience 20 and a factor of 0.5. This means that the learning rate will be multiplied by 0.5 when the RMSE in the early stopping set does not decrease in 20 epochs. By doing this, the training set is prevented from becoming stuck in the local optima (Smith and Topin, 2018).

In this study, 10-fold walk-forward validation is used with a sliding window of 1. This means that every time the training data fold increases, it will increase by 1. This is adjusted to what happens on the stock market, where stock data will increase by 1 every day. The RMSE of each fold will be averaged to obtain a validation error that will be used to update the surrogate model on the BO.

Table 4
Hyperparameters to be optimized along with their limitations.

Hyperparameter	Interval
Number of LSTM layers	[1–3]
Number of units per layer	[100−500]
Dropout rate per layer	[0–0.5]
Batch Size	[1–64]
Optimizer	[Adam, RMSprop]
Learning Rate	[0.01, 0.00001]
Timestamps	[5, 10, 20, 30]

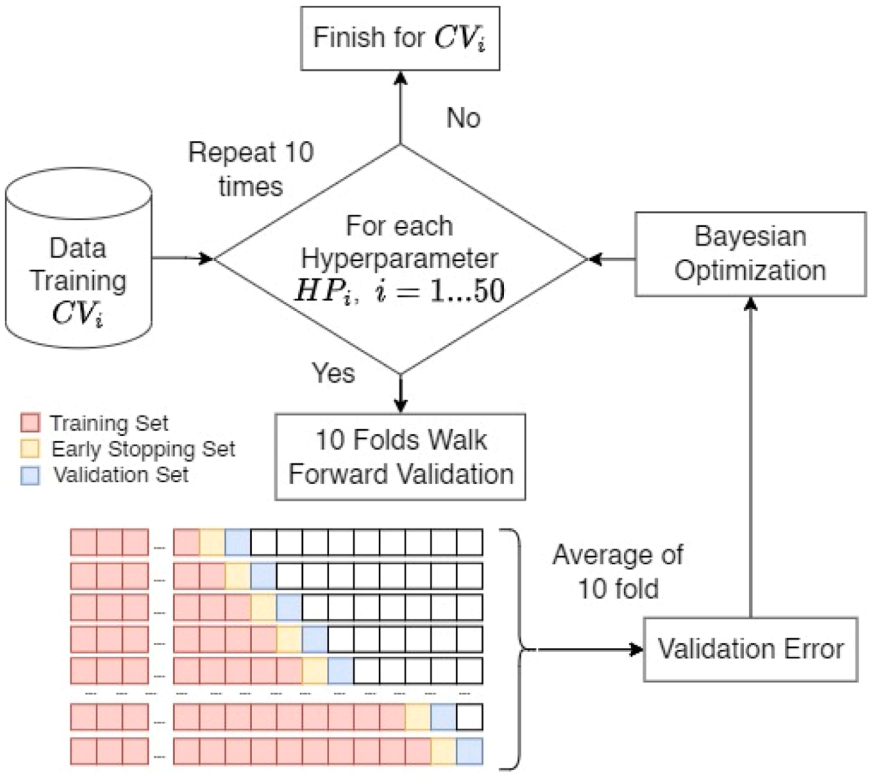


Fig. 3. HPO Process and Walk Forward Validation.

3.5. Comparing variable combinations

To get a better RMSE estimate, each of the best models will be retrained with 60-fold walk-forward validation so that they can get model performance in 60 days or about 3 months. The process will be run 10 times to see the stability of each model. The average RMSE obtained from the 60-fold will be compared to determine the best combination of variables. The model with the best combination will be used to predict the testing data to obtain an error estimate from the prediction. The results obtained will also be compared with the errors obtained from the research model conducted by Heiden and Parpinelli (2021) and Mndawe et al. (2022).

4. Results and experiments

4.1. Hyperparameter optimization

There are 13,406 news contain the words "Bank Rakyat Indonesia", "BRI", or "BBRI", of which 4256 are from *detik finance* and 9150 are from Bisnis.com. Fig. 4 shows the sentiment of news about BRI tends to be positive and only about 7.89 % or 1056 articles have a sentiment score less than 0. The sentiment score will be used as a variable and used in modeling.

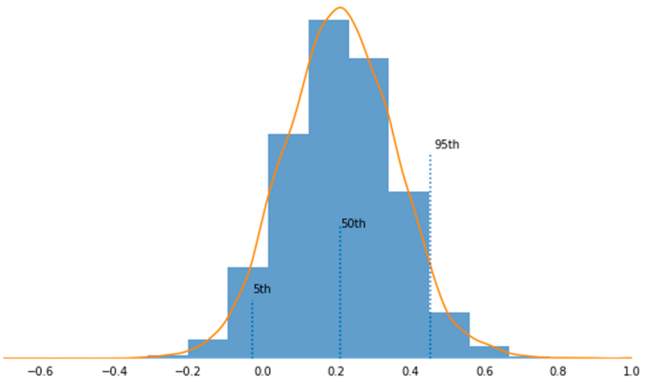


Fig. 4. Score sentiment histogram.

For each combination of variables, Bayesian Optimization is run 10 times where each run will evaluate 50 different hyperparameters. So, for one combination of variables, 500 different hyperparameters are evaluated. There are a total of 4000 distinct hyperparameters due to the 8 variable combinations. Each hyperparameter is executed 10-fold, so a total of 40,000 models have been run.

Every time Bayesian Optimization is run, it will produce the best different hyperparameters. In other words, iterations 1–10 on each combination of variables will produce different hyperparameters. So that each iteration will produce a different RMSE as well. For all iterations, the average and standard deviation will be calculated. The smaller the standard deviation, the more stable the Bayesian Optimization algorithm is. This is because the smaller the standard deviation means the performance of each iteration is almost the same.

According to Table 5, the third column shows that the standard deviation of Bayesian Optimization is in the range of 1.32–5.53. When indicator variables are used, Bayesian Optimization is increasingly

Table 5
Average and standard deviation RMSE of 10 iterations of Bayesian optimization.

VC	Average RMSE	Standard Deviation RMSE	There are Technical Indicators or not
VC_1	40.85	1.32	No
VC_2	39.64	2.38	No
VC_3	39.97	2.21	No
VC_4	43.83	5.53	Yes
VC_5	**38.66**	**1.63**	No
VC_6	48.36	4.82	Yes
VC_7	45.16	2.50	Yes
VC_8	45.28	4.65	Yes

unstable, which can be seen from the larger standard deviation when compared to a combination of variables that do not use technical indicators. The smallest average is VC_5 (History of closing price, sentiment score, and rupiah exchange rate) with a small standard deviation of 1.63.

To see if there are differences in RMSE values between variable combinations, the Kruskal-Wallis test is performed. H_0 of the test is all combinations of variables have the same RMSE distribution. The result obtained is the H statistic of 38.80 with a p-value of 2.13×10^{-6}. Because the p-value is smaller than 5 %, the decision taken is to reject H_0, which means that statistically there is not enough evidence to say that every combination of variables has the same RMSE distribution. To see which combinations of variables are different, a post hoc test was performed using the Mann-Whitney test.

Based on Table 6, for the combination of variables that have technical indicators (VC_4, VC_6, VC_7, VC_8), when tested with each other using Mann Whitney, it will result in failure to reject H_0, which means that statistically there is not enough evidence to say that the RMSE distribution in the four combinations of variables is different. However, when the four combinations of variables were tested with a combination of variables that had no technical indicators, it resulted in a rejection of H_0, except for VC_1 dan VC_4. This shows that, statistically, Bayesian Optimization produces the same RMSE distribution between the combinations of variables with technical indicators, but the RMSE value is still greater than those without technical indicators.

Table 7 is a list of the best hyperparameters for each variable combination. There is a pattern that combinations of variables that contain technical indicators (VC_4, VC_6 , VC_7, VC_8) have 2 or 3 LSTM layers, while those without technical indicators are enough with just 1 LSTM layer. This shows that the more variables used, the more the model used requires a complex architecture to be able to understand the pattern of the data.

It can be seen that each variable combination has a different hyperparameter configuration to achieve the minimum error which confirms what Kohavi and John said (Kohavi and John, 1995) that different configurations of hyperparameters tend to be good for different data sets. Therefore, it is very important to do HPO before determining the best combination of variables so that the error being compared is the minimum error. If you don't do HPO then the conclusions drawn must refer to the hyperparameters used and cannot be generalized in general to a particular model.

4.2. Best variable combination

As input for the time series, various approaches can be employed, such as incorporating lagged variables (Han et al., 2025) or selecting the optimal combination of variables. For each combination of variables, the best hyperparameters are listed in Table 7. These variables are retrained on 60-fold walk-forward validation. For each combination of variables, modeling is done 10 times so that it will produce 10 different RMSE and then the average and standard deviation are calculated. In this study, the standard deviation describes how stable the model is for randomness.

According to the result presented in Table 8, VC_1 is a variable with the smallest standard deviation of the RMSE 60-fold walk-forward validation of the best hyperparameters. This shows that in each iteration the performance of the model on VC_1 is almost similar. However, the average RMSE value is very high, which is 137.81, even close to the RMSE value from a combination of variables using technical indicators. This indicates that only using the history of stock prices is not enough to make the model better. The model needs other variables to better understand the pattern from the data.

It can be revealed from Table 8 that the VC_2 (stock price history and sentiment score) is the best model for predicting BRI stock prices. The model performed well even though it was tested at 60-fold. Of the 60-folds, there are only 17-folds whose RMSE values are more than 100 and 5-folds that are more than 150. The lowest RMSE in the 1st fold is 25.20 and the highest RMSE is in the 45th fold, which is 201.42. Fig. 5 below is an illustration of the architecture and hyperparameters of the best model obtained (VC_2).

Furthermore, it will be seen the percentage decrease in RMSE between the combination of variables. To see it will use the formula that has been used in the research (Siami-Namini et al., 2018). The following formula is used.

$$\%Changes = \frac{RMSE_2 - RMSE_1}{RMSE_1} \times 100 \quad (1)$$

Based on Table 8 and the calculation using formula (1), the addition of sentiment scores in the VC_2 model can reduce RMSE by 39.55 % when compared to the VC_1 model which only uses stock price history. It's RMSE changed from 137.81 (RMSE VC_1) to 83.31 (RMSE VC_2). However, when the rupiah exchange rate was added so that the model used 3 variables (stock price history, sentiment score, and rupiah exchange rate), the RMSE increased by 4.73 %, from 83.31 (RMSE VC_2) to 87.26 (RMSE VC_5). Even worse, when the model that uses 3 variables is added with 10 technical indicators so that it uses 13 variables, the RMSE increases by 50.59 %, from 87.26 (RMSE VC_5) to 131.38 (RMSE VC_8).

4.3. Evaluation of the best model

Based on the HPO process, it is found that the VC_2 model is the best model to predict BRI's stock price. The model will then be evaluated on the testing data to get an estimate of the prediction error. In addition, the LSTM model architecture from other studies will be used as a

Table 6
A p-value of the Mann-Whitney test for each pair of variable combinations.

	VC_1	VC_2	VC_3	VC_4	VC_5	VC_6	VC_7	VC_8
VC_1	1.000	0.307	0.358	0.089	0.011	0.002	0.002	0.026
VC_2	0.307	1.000	1.000	0.031	0.212	0.001	0.001	0.009
VC_3	0.358	1.000	1.000	0.045	0.140	0.001	0.001	0.014
VC_4	0.089	0.031	0.045	1.000	0.011	0.076	0.571	0.473
VC_5	0.011	0.212	0.140	0.011	1.000	0.001	0.000	0.002
VC_6	0.002	0.001	0.001	0.076	0.001	1.00	0.054	0,241
VC_7	0.002	0.001	0.001	0.571	0.000	0.054	1.000	0.791
VC_8	0.026	0.009	0.014	0.473	0.002	0,241	0.791	1.000

The blue box identifies the combination of variables there is no technical indicator, the green color has a technical indicator and orange means the p-value is less than 5 %

Table 7
The best hyperparameters for each variable combination.

VC	Hyperparameter					
	Unit Layer	Dropout Rate	Batch Size	Optimizer	Learning Rate	Timestamps
VC_1	[150]	[0.5]	32	RMSprop	0.00001	30
VC_2	[250]	[0.48]	16	Adam	0.00375	30
VC_3	[100]	[0.41]	32	RMSprop	0.00684	20
VC_4	[150,250]	[0.46, 0.05]	32	Adam	0009940	5
VC_5	[300]	[0.37]	48	RMSprop	0.00559	10
VC_6	[150, 100, 100]	[0.23, 0.00, 0.00]	32	Adam	0.00586	30
VC_7	[500, 150, 150]	[0.31, 0.06, 0.49]	64	RMSprop	0.00712	30
VC_8	[500, 350, 400]	[0.29, 0.01, 0.00]	32	Adam	0.00740	5

Table 8
The mean and standard deviation of the RMSE 60-fold walk-forward validation of the best hyperparameters for each combination of variables.

VC	Average RMSE	Standard Deviation RMSE	There are Technical Indicators or not
VC_1	137.81	2.63	No
VC_2	**83.31**	**3.82**	**No**
VC_3	91.02	2.72	No
VC_4	134.37	7.30	Yes
VC_5	87.26	3.36	No
VC_6	141.02	6.19	Yes
VC_7	160.38	11.53	Yes
VC_8	131.38	7.44	Yes

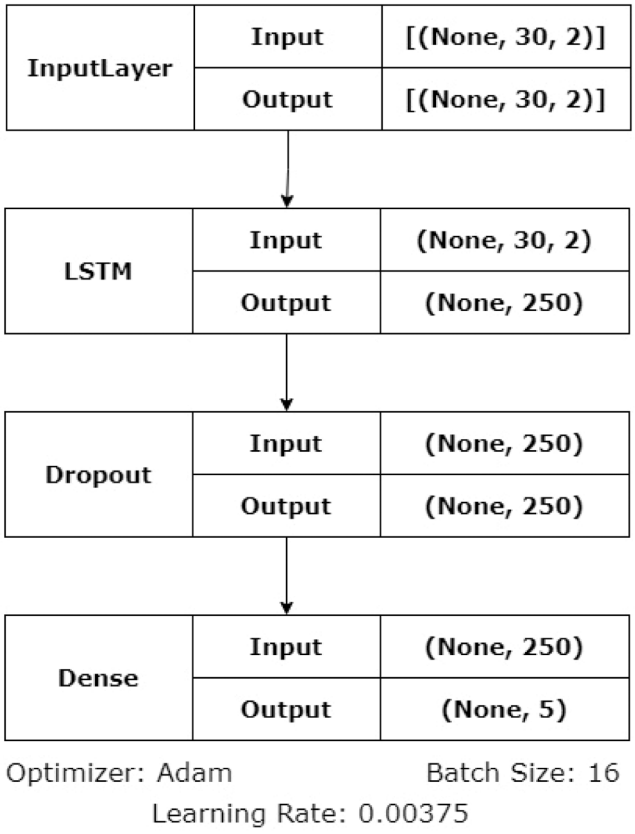


Fig. 5. Illustration of the architecture and hyperparameters of the VC_2 model.

comparison.

Not all studies describe the architecture completely and it is very difficult to get the source code of the model. Therefore, only 2 studies will be selected that will be used as comparisons, there is the model in the study (Heiden and Parpinelli, 2021; Mndawe et al., 2022). The two studies were chosen because the architecture of the model used is described in the paper, although the hyperparameter configuration is not fully described. The thing that will be changed is the size of the input and output layers. The input layer will be adjusted so that it can accept 2 variables, namely stock price history and score sentiment. The output layer is changed so that it consists of 5 units because the modeling in this study predicts the next 5 days directly.

The research (Heiden and Parpinelli, 2021) uses a model that has 2 LSTM layers, with 64 units of the first layer and 32 units of the second layer. Both LSTM layers use hyperbolic tangent (tanh) as their activation function. Meanwhile, for the output layer, the activation function of the Rectified Linear Unit (ReLU) is used. The model uses 60 timestamps and 100 epochs. Because the model uses a fixed number of epochs, it means that the training process will not use early stopping. In addition, the model does not use dropout layers. In this study, it was not stated how many batch sizes, optimizers, and learning rates were used. Therefore, the batch size, optimizer, and learning rate will be used as in VC_2.

The study (Mndawe et al., 2022) conducted 3 different experiments. The model chosen to be used in this study is the best model in the second experiment in the second part. The experiment was chosen because it is closer to what was done in this study. Mndawe et al. (2022) used many variables (multivariate) to predict the closing price of Vodacom shares. The model has 1 layer LSTM with 200 units. The model uses 20 timestamps and is trained in 100 epochs using a learning rate of $1e^{-4}$. Early stopping and a dropout layer are not used in the modeling process. The research also doesn't mention batch size and optimizer so it will use the batch size and optimizer from the VC_2 model. In addition, the activation function used in each layer is not stated so the activation function used by the VC_2 model.

The model evaluation process will be the same as the HPO process using walk-forward validation but with 5 different scenarios. The data from January to March 2022 consists of 57 observations, so a 57-fold walk-forward validation will be carried out. In the first scenario, the model will be trained every day, so it will be the same as the walk-forward validation process used during the hyperparameter optimization process. In the second scenario, the model will be trained every 5 days or 1 week. For example, the model is trained on Monday, then the model will be used on Monday, Tuesday, Wednesday, Thursday, and Friday, then on the following Monday, the model will be trained again with new data. The third scenario will be trained once every 10 days or 2 weeks, the fourth scenario will be trained once every 15 days or 3 weeks and the fifth scenario will be trained every 20 days or 1 month. Each scenario will be repeated 10 times and then the average and standard deviation of the RMSE are calculated. The calculated standard deviation indicates the model's stability level.

According to Table 9, it can be seen that in general, for 57-fold walk-forward validation, the VC_2 model has the best performance in almost all scenarios. Except for the 5th scenario, the RMSE of the VC_2 model is only 0.30 greater than the model (Mndawe et al., 2022). The RMSE of the model (Heiden and Parpinelli, 2021) is very different from the other two models. If explored further, it turns out that the prediction of the model is not very good as shown in Fig. 6. This is because the architecture used cannot capture patterns from the data, perhaps due to the

Table 9
The mean and standard deviation of 10 iterations for each scenario.

Scenario	Average RMSE (±Standard Deviation)		
	Model VC_2	**Model** (Heiden and Parpinelli, 2021)	**Model** (Mndawe et al., 2022)
Trained every day	**96.61**(±**2.75**)	4180.63(±49.23)	118.77(±1.30)
Trained every 5 days	102.51(±7.26)	4215.80(±121.25)	119.16(±2.46)
Trained every 10 days	105.48(±9.21)	4276.47(±0.00)	116.39(±3.71)
Trained every 15 days	106.89(±10.12)	4236.28(±127.10)	124.57(±4.09)
Trained every 20 days	126.30(±10.03)	4202.32(±234.49)	126.00(±5.81)

use of inappropriate activation functions.

It is also seen that the less often the model is trained, the lower the performance of the model, this is clearly illustrated in Fig. 6. This shows that it is very important to train the model as often as possible. In the case of stock predictions where the data is always increasing every day, it is very good if the model is always trained every day. Therefore, the use of the walk-forward validation method when evaluating the time series model is the right choice because the method will be similar to the actual conditions where the data always increases with time.

The MAE of 10 iterations of the 57-fold walk forward validation model VC_2 which is trained daily is 86.97 ± 2.51. When viewed per fold (Fig. 7), the lowest MAE is 20.98 (fold 2), which means that the prediction of BRI's stock price on that day is only 20.98 different from the actual stock price. At fold 21–26, the performance of the model is very poor, with the highest MAE value at fold 24 which reaches 343.30. This is because the model failed to predict a high increase in BRI's stock price at the beginning of February 2022.

Because the model making is a multi-step model, it can be calculated how the model's performance is when predicting one day ahead, two days ahead, and up to 5 days ahead. From Table 10, the closer the prediction day, the better the model performance. Therefore, the closer the day of prediction, the more reliable the prediction results. In addition, predictions for the next few days can be used as an early warning.

Fig. 8 is a line chart from the first to fifth-day predictions. The farther the prediction day, the poorer the model performance where the difference between the predicted value and the actual value is getting bigger.

In addition, based on Table 10, it is also seen that the average value is higher than the median value, so the MAE distribution is skewed to the right. This shows that there is an outlier value where on a certain date the model has poor performance. The daily MAE distribution is visible in the boxplot of Fig. 9.

5. Discussion

Each combination of variables is performed 10 times with Bayesian Optimization so that there are 500 different hyperparameters that are evaluated. Table 11 shows how many trials the lowest RMSE was

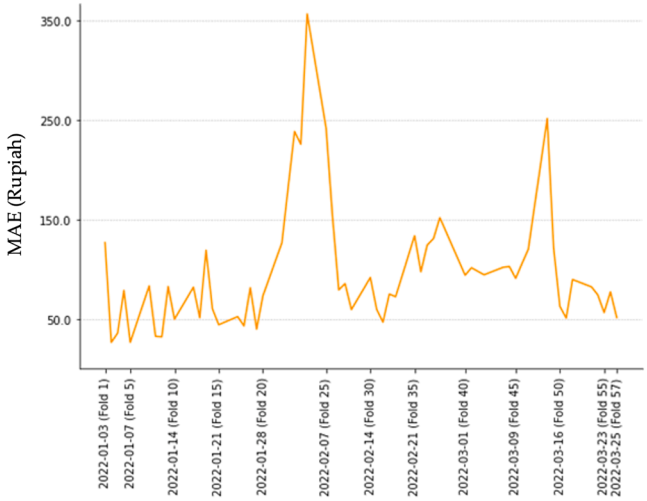


Fig. 7. MAE per fold of VC_2 model trained daily.

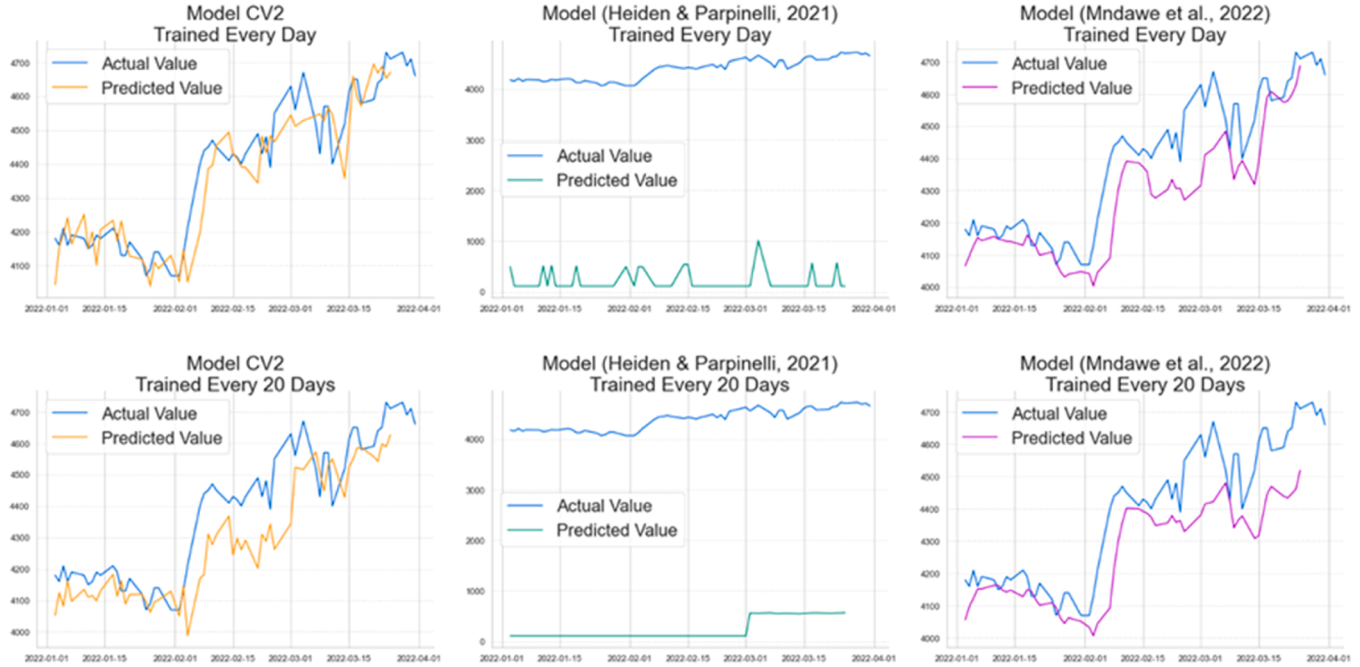


Fig. 6. Real value vs. prediction (first day) of BRI stock price using a model that is trained every 1 day and 20 days.

Table 10
MAE Model VC_2 for predictions from the first to the fifth day.

	MAE Prediction Day To-				
	1	2	3	4	5
Average	70.16	78.25	88.18	95.48	102.82
Standard Deviation	44.36	63.20	72.14	82.53	85.63
Minimum	11.06	10.27	9.35	18.47	9.48
First Quartile	38.24	43.82	43.76	48.12	44.60
Median	55.74	58.47	66.57	72.33	75.50
Third Quartile	92.69	89.07	105.46	94.93	131.61
Maximum	202.83	351.07	390.93	402.94	413.32

obtained. For VC_2, it took 382 trials to get the best hyperparameter, which is the best hyperparameter in this study. For other variable combinations, it also requires more than 100 experiments to obtain the best hyperparameter. This shows the importance of doing hyperparameter optimization with many experiments.

The more experiments that are done, the greater the chance to find the most optimal hyperparameter. Table 11 also shows that there is no best hyperparameter obtained directly in the first experiment. Other studies without performing hyperparameters as presented in Heiden and Parpinelli (2021) and Mndawe et al. (2022) is very possible that the calculated model does not yet have the best hyperparameters. This is also evident from the experimental results when the models presented in Heiden and Parpinelli (2021) and Mndawe et al. (2022) are compared with the VC_2 model obtained from the hyperparameter optimization process, the VC_2 model is better than the models from the two studies.

Nine out of the 10 technical indicators used in this study were also used in research (Kara et al., 2011; Sivapurapu, 2020). However, these two studies do not explain further the influence of these technical indicators. Meanwhile, Table 8 shows that for the combination of variables that contain technical indicators, the RMSE value is quite high, which is above 130. This shows that the use of technical indicators does not make the model better.

This is different from the results of research (Neely et al., 2014; Mohapatra et al., 2022) which state that the use of technical indicators has a significant impact on the modeling. The technical indicators used in the two studies are indeed different from those used in this study. This shows that not all technical indicators have a bad impact on modeling. It is possible that the 10 technical indicators used in this study are not good for capturing trend developments in BRI stocks. Or maybe it's not the right choice to combine the 10 technical. Maybe only using 2 or 3 technical indicators will make the model better.

In addition, after further review, it turns out that several technical indicators have high correlation values with each other. For example, for technical indicators, SMA and WMA have a very high correlation,

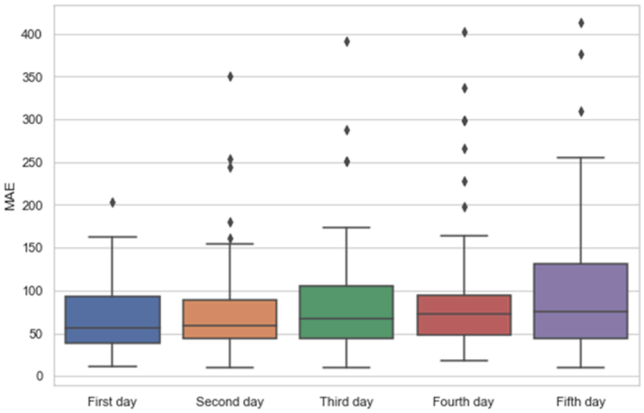


Fig. 9. Boxplot MAE Model VC_2 for the prediction of the first to fifth day.

Table 11
Number of Trials When RMSE is Lowest.

VC	Trial Order to
VC_1	433
VC_2	382
VC_3	110
VC_4	216
VC_5	499
VC_6	250
VC_7	169
VC_8	335

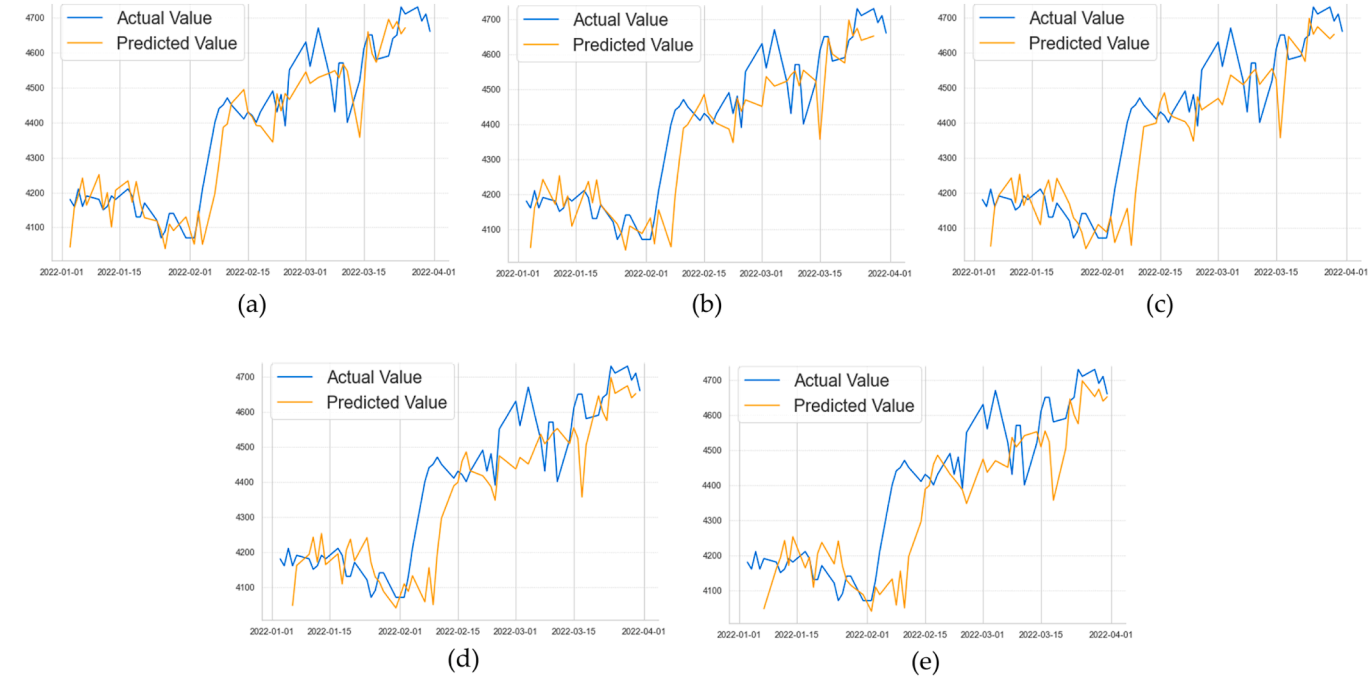


Fig. 8. The predicted value of the daily trained VC_2 model for (a) first day; (b) second day; (c) third day; (d) fourth day; (e) fifth day.

namely 0.99. This shows the occurrence of multicollinearity in the independent variables. This multicollinearity problem should have been solved at the data preprocessing stage as was done in the research (Kumar et al., 2022). The study checked multicollinearity in the data before modeling using LSTM. The existence of independent variables that have a high correlation makes the model use variables that contain almost the same information. Of course, this does not improve the model, on the contrary, it can make it harder for the model to identify trends in the data. This multicollinearity may also be the cause of the 10 technical indicators that are used which make the model worse.

The best model obtained in this study uses stock price history variables and sentiment scores. This is the same as the results in research (Heiden and Parpinelli, 2021; Mndawe et al., 2022) which show that adding sentiment scores can improve the performance of the model. However, both studies did not use walk-forward validation in the modeling process. So, the environment when the modeling process is different from the actual stock market environment. While in this study, the validation method has used walk-forward validation so that the environment in the training process is the same as the actual conditions in the stock market. This makes our results more reliable than the results obtained in the two previous studies.

6. Conclusions

The purpose of this study is to present a general framework for conducting time series modeling. The results obtained show that each combination of variables has the best hyperparameters respectively. Therefore, it is necessary to perform hyperparameter optimization for each combination of variables. The validation method that we recommend is to use walk-forward validation with a sliding window adjusting to actual conditions. Because the stock data increases by 1 every day, the sliding window used is 1. In the hyperparameter optimization process, we recommend using a smaller number of folds so that we can save on the computational process needed and focus more on trying out many hyperparameters. After each combination of variables has the best hyperparameter, then it is retrained in a larger number of folds to determine the best combination of variables.

In this study, the best model for predicting BRI stock prices is a model with a combination of stock price history variables and sentiment scores. The combination of these variables cannot be generalized to other stocks. This is the limitation of this research. Future research can analyze using more than 1 stock so that the results obtained can be generalized to other stocks. In addition, it can also determine which variables are most influential in predicting stock prices and answer the question of whether, in general, sentiment score is a variable that has a significant effect on the overall stock price.

In addition, the limitation of this study is the use of 10 technical indicators that are used in one group. Therefore, we cannot conclude how the influence of each technical indicator on the predictions made. In this study, only 10 of the hundreds of technical indicators exist. In addition, it is also necessary to research which technical indicators have a good impact on modeling and vice versa.

The sentiment model and stock prediction that we use are also limitations in this study. The sentiment model that we use is only a lexical-based (VADER) model, which of course is not very good when compared to machine learning-based models. The prediction model used is only LSTM, even though there are many other deep learning models such as Bidirectional LSTM, LSTM Encoder-Decode, Convolutional Neural Network (CNN), and so on. The use of many and varied models can be a study that needs to be done in the future so that it can enrich the insights that will be obtained later.

Ethics approval

Not applicable; we are utilizing secondary data for our research, which has been previously collected and does not involve any direct interaction with human subjects.

Informed Consent Statement

Not applicable.

Informed consent

Not Applicable; are utilizing secondary data for our research, which has been previously collected and does not involve any direct interaction with human subjects.

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CRediT authorship contribution statement

Eko Putra Wahyuddin: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Rezzy Eko Caraka**: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Robert Kurniawan**: Writing – review & editing, Writing – original draft. **Wahyu Caesarendra**: Writing – review & editing. **Prana Ugiana Gio**: Writing – review & editing. **Bens Pardamean**: Writing – review & editing.

Declaration of Competing Interest

The authors have no competing interests to declare that are relevant to the content of this article. Data is available upon request.

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Data availability

The source code and the material and findings data of this study are openly available in full access by GitHub: https://github.com/Rezzy94

References

Ardyanta, E.I., Sari, H., 2021. A prediction of stock price movements using support vector machines in Indonesia. J. Asian Financ., Econ. Bus. 8 (8), 399–407. https://doi.org/10.13106/JAFEB.2021.VOL8.NO8.0399.

Ariyo, A.A., Adewumi, A.O., & Ayo, C.K. (2014). Stock Price Prediction Using the ARIMA Model. 2014 UKSim-AMSS 16th International Conference on Computer Modelling and Simulation, 106–112. https://doi.org/10.1109/UKSim.2014.67.

Aslam, S., Rasool, A., Wu, H. and Li, X. (2024), "CEL: A Continual Learning Model for Disease Outbreak Prediction by Leveraging Domain Adaptation via Elastic Weight Consolidation", 15 January, doi: 10.1101/2024.01.13.575497.

Barra, S., Carta, S.M., Corriga, A., Podda, A.S., Recupero, D.R., 2020. Deep learning and time series-to-image encoding for financial forecasting. IEEE/CAA J. Autom. Sin. 7 (3), 683–692. https://doi.org/10.1109/JAS.2020.1003132.

Bergmeir, C., Benítez, J.M., 2012. On the use of cross-validation for time series predictor evaluation. Inf. Sci. 191, 192–213. https://doi.org/10.1016/j.ins.2011.12.028.

Bergstra, J., & Bengio, Y. (2012). Random Search for Hyper-Parameter Optimization. 25.

Biswas, S., 2020. Scope of sentiment analysis on news articles regarding stock market and GDP in struggling economic condition. Int. J. Emerg. Trends Eng. Res. 8 (7), 3594–3609. https://doi.org/10.30534/ijeter/2020/117872020.

Börjesson, L., Singull, M., 2020. Forecasting financial time series through causal and dilated convolutional neural networks. Entropy 22 (10), 1094. https://doi.org/10.3390/e22101094.

Brownlee, J. (2018). Deep Learning for Time Series Forecasting: Predict the Future with MLPs, CNNs and LSTMs in Python. Machine Learning Mastery. https://books.google.co.id/books?id=o5qnDwAAQBAJ.

Bustos, O., Pomares-Quimbaya, A., 2020. Stock market movement forecast: A Systematic review. Expert Syst. Appl. 156, 113464. https://doi.org/10.1016/j.eswa.2020.113464.

Caraka, R.E., Supardi, K., Kurniawan, R., Kim, Y., Gio, P.U., Yuniarto, B., Mubarok, F.Z., et al., 2024. Empowering deaf communication: a novel LSTM model for recognizing Indonesian sign language. Universal Access in the Information Society. Springer Science and Business Media Deutschland GmbH. https://doi.org/10.1007/s10209-024-01095-1.

Chen, W., Jiang, Q., Jia, X., Rasool, A., Jiang, W., 2023. A high-frequency stock price prediction method based on mode decomposition and deep learning (CCIS). Commun. Comput. Inf. Sci. 1796. https://doi.org/10.1007/978-981-99-3300-6_5.

Chen, Y., Rao, M., Feng, K., Zuo, M.J., 2022. Physics-Informed LSTM hyperparameters selection for gearbox fault detection. Mech. Syst. Signal Process. 171. https://doi.org/10.1016/j.ymssp.2022.108907.

Chen, Y., Niu, G., Li, Y., Li, Y., 2023. A modified bidirectional long short-term memory neural network for rail vehicle suspension fault detection. Veh. Syst. Dyn. 61 (12), 3136–3160. https://doi.org/10.1080/00423114.2022.2158879.

Deng, S., Mitsubuchi, T., Shioda, K., Shimada, T., & Sakurai, A. (2011). Combining Technical Analysis with Sentiment Analysis for Stock Price Prediction. 2011 IEEE Ninth International Conference on Dependable, Autonomic and Secure Computing, 800–807. https://doi.org/10.1109/DASC.2011.138.

Derakhshan, A., Beigy, H., 2019. Sentiment analysis on stock social media for stock price movement prediction. Eng. Appl. Artif. Intell. 85, 569–578. https://doi.org/10.1016/j.engappai.2019.07.002.

Graves, A., Liwicki, M., Fernandez, S., Bertolami, R., Bunke, H., Schmidhuber, J., 2009. A novel connectionist system for unconstrained handwriting recognition. IEEE Trans. Pattern Anal. Mach. Intell. 31 (5), 855–868. https://doi.org/10.1109/TPAMI.2008.137.

Han, P., Chen, H., Rasool, A., Jiang, Q., Yang, M., 2025. MFB: a generalized multimodal fusion approach for bitcoin price prediction using time-lagged sentiment and indicator features. Expert Syst. Appl. 261. https://doi.org/10.1016/j.eswa.2024.125515.

Heiden, A., Parpinelli, R.S., 2021. Applying LSTM for Stock Price Prediction with Sentiment Analysis. An. Do 15. Congr. Bras. De. Inteligência Comput. 1–8. https://doi.org/10.21528/CBIC2021-45.

Hochreiter, S., Schmidhuber, J., 1997. Long short-term memory. Neural Comput. 9 (8), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735.

Huang, B., Ding, Q., Sun, G., Li, H., 2018. Stock prediction based on Bayesian-LSTM. Proc. 2018 10th Int. Conf. Mach. Learn. Comput. 128–133. https://doi.org/10.1145/3195106.3195170.

Hutto, C., Gilbert, E., 2014. VADER: a parsimonious rule-based model for sentiment analysis of social media text. Proc. Int. AAAI Conf. Web Soc. Media 8 (1), 216–225.

Ji, Y., Huang, Y., Zeng, J., Ren, L., Chen, Y., 2025. A physical-data-driven combined strategy for load identification of tire type rail transit vehicle. Reliab. Eng. Syst. Saf. 253. https://doi.org/10.1016/j.ress.2024.110493.

Jiang, W., 2021. Applications of deep learning in stock market prediction: Recent progress. Expert Syst. Appl. 184, 115537. https://doi.org/10.1016/j.eswa.2021.115537.

Kara, Y., Acar Boyacioglu, M., Baykan, Ö.K., 2011. Predicting direction of stock price index movement using artificial neural networks and support vector machines: The sample of the Istanbul Stock Exchange. Expert Syst. Appl. 38 (5), 5311–5319. https://doi.org/10.1016/j.eswa.2010.10.027.

Kohavi, R., John, G.H., 1995. Automatic parameter selection by minimizing estimated error. Machine Learning Proceedings 1995. Elsevier, pp. 304–312. https://doi.org/10.1016/B978-1-55860-377-6.50045-1.

Kouassi, K.H., & Moodley, D. (2020a). An analysis of deep neural networks for predicting trends in time series data. https://doi.org/10.48550/ARXIV.2009.07943.

Kouassi, K.H., & Moodley, D. (2020b). Automatic deep learning for trend prediction in time series data. https://doi.org/10.48550/ARXIV.2009.08510.

Kuhn, M., Johnson, K., 2020. Feature Engineering and Selection: A Practical Approach for Predictive Models. CRC Press, Taylor & Francis Group.

Kumar, V.K., Kv, R., Rakesh, V., 2022. Optimizing Lstm and Bi-Lstm models for multivariate time series prediction and performance comparison with classical statistical models. SSRN Electron. J. https://doi.org/10.2139/ssrn.4129886.

Le, Q.V., Jaitly, N., & Hinton, G.E. (2015). A Simple Way to Initialize Recurrent Networks of Rectified Linear Units. *ArXiv:1504.00941 [Cs]*. http://arxiv.org/abs/1504.00941.

Mndawe, S.T., Paul, B.S., Doorsamy, W., 2022. Development of a Stock Price Prediction Framework for Intelligent Media and Technical Analysis. Appl. Sci. 12 (2), 719. https://doi.org/10.3390/app12020719.

Mohan, S., Mullapudi, S., Sammeta, S., Vijayvergia, P., & Anastasiu, D.C. (2019). Stock Price Prediction Using News Sentiment Analysis. 2019 IEEE Fifth International Conference on Big Data Computing Service and Applications (BigDataService), 205–208. https://doi.org/10.1109/BigDataService.2019.00035.

Mohapatra, S., Mukherjee, R., Roy, A., Sengupta, A., Puniyani, A., 2022. Can Ensemble Machine Learning Methods Predict Stock Returns for Indian Banks Using Technical Indicators? J. Risk Financ. Manag. 15 (8), 350. https://doi.org/10.3390/jrfm15080350.

Mustapa, F.H., Ismail, M.T., 2019. Modelling and forecasting S&P 500 stock prices using hybrid Arima-Garch Model. J. Phys.: Conf. Ser. 1366 (1), 012130. https://doi.org/10.1088/1742-6596/1366/1/012130.

Neely, C.J., Rapach, D.E., Tu, J., Zhou, G., 2014. Forecasting the equity risk premium: the role of technical indicators. Manag. Sci. 60 (7), 1772–1791.

Nguyen, T.H., Shirai, K., Velcin, J., 2015. Sentiment analysis on social media for stock movement prediction. Expert Syst. Appl. 42 (24), 9603–9611. https://doi.org/10.1016/j.eswa.2015.07.052.

Nti, I.K., Adekoya, A.F., Weyori, B.A., 2020. Predicting stock market price movement using sentiment analysis: evidence From Ghana. Appl. Comput. Syst. 25 (1), 33–42. https://doi.org/10.2478/acss-2020-0004.

Prechelt, L., 1998. Early stopping—but when? In: Orr, G.B., Müller, K.-R. (Eds.), Neural Networks: Tricks of the Trade, 1524 Springer Berlin Heidelberg, pp. 55–69. https://doi.org/10.1007/3-540-49430-8_3.

Ruan, Y., Durresi, A., Alfantoukh, L., 2018. Using Twitter trust network for stock market analysis. Knowl. -Based Syst. 145, 207–218. https://doi.org/10.1016/j.knosys.2018.01.016.

Siami-Namini, S., Tavakoli, N., Siami Namin, A., 2018. A Comparison of ARIMA and LSTM in Forecasting Time Series. 17th IEEE Int. Conf. Mach. Learn. Appl. (ICMLA) 2018, 1394–1401. https://doi.org/10.1109/ICMLA.2018.00227.

Sivapurapu, S.A., 2020. Comparitive study of time series and deep learning algorithms for stock price prediction. Int. J. Adv. Comput. Sci. Appl. 11 (6). https://doi.org/10.14569/IJACSA.2020.0110658.

Smith, L.N., & Topin, N. (2018). Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates. *ArXiv:1708.07120 [Cs, Stat]*. http://arxiv.org/abs/1708.07120.

Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: A simple way to prevent neural networks from overfitting. J. Mach. Learn. Res. 15 (1), 1929–1958.

Tan, T.Z., Quek, C., Ng, G.S., 2007. Biological brain-inspired genetic complementary learning for stock market and bank failure prediction. Comput. Intell. 23 (2), 236–261. https://doi.org/10.1111/j.1467-8640.2007.00303.x.

Toharudin, T., Pontoh, R.S., Caraka, R.E., Zahroh, S., Lee, Y., Chen, R.C., 2021. Employing Long Short-Term Memory and Facebook Prophet Model in Air Temperature Forecasting (Vol. early acces). Commun. Stat. - Simul. Comput. 1–12. https://doi.org/10.1080/03610918.2020.1854302.

Torres, J.F., Hadjout, D., Sebaa, A., Martínez-Álvarez, F., Troncoso, A., 2021. Deep learning for time series forecasting: a survey. Big Data 9 (1), 3–21. https://doi.org/10.1089/big.2020.0159.

Vargas, M.R., dos Anjos, C.E.M., Bichara, G.L.G., & Evsukoff, A.G. (2018). Deep Leaming for Stock Market Prediction Using Technical Indicators and Financial News Articles. 2018 International Joint Conference on Neural Networks (IJCNN), 1–8. https://doi.org/10.1109/IJCNN.2018.8489208.

Victoria, A.H., Maragatham, G., 2021. Automatic tuning of hyperparameters using Bayesian optimization. Evol. Syst. 12 (1), 217–223. https://doi.org/10.1007/s12530-020-09345-2.

Wu, S., Liu, Y., Zou, Z., Weng, T.-H., 2022. S_I_LSTM: Stock price prediction based on multiple data sources and sentiment analysis. Connect. Sci. 34 (1), 44–62. https://doi.org/10.1080/09540091.2021.1940101.

Yu, T., & Zhu, H. (2020). Hyper-Parameter Optimization: A Review of Algorithms and Applications. https://doi.org/10.48550/ARXIV.2003.05689.

Zheng, A., Casari, A., 2018. Feature Engineering for Machine Learning: Principles and Techniques for Data Scientists, First edition. O'Reilly.

Zhong, X., Enke, D., 2017. A comprehensive cluster and classification mining procedure for daily stock market return forecasting. Neurocomputing 267, 152–168. https://doi.org/10.1016/j.neucom.2017.06.010.